

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **DotNet Technology**

Code No: **CACS302** Semester: **V**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What do you mean by "safe" programming language? Explain any four applied technologies in DotNet framework.
3. List any five canonical keywords in "C#". Write a C# program to initialize and display array element with sum of each power.
4. What is string interpolation? How passing argument by value is differing from passing argument by reference? Explain with program.
5. What are sealed classes in C#? Write a C# program to illustrate the concept of abstract class and abstract method.
6. What is virtual method? Write a program to illustrate multi-level inheritance with virtual method.
7. What is different between index and properties? Explain the concept of generic class in C# with 8 suitable program.
8. Write a program to create form for calculating simple interest in one ASP.NET page and display the simple interest in another page of ASP.NET.

Group C

Attempt any TWO questions[2x10=20]

9. a) What is System Exception? Write a C# program that will read balance and withdraw amount from keyboard and display the remaining balance on screen. if balance is greater than withdrawal amount otherwise throw an Application Exception with appropriate message. b) List any five LINQ standard operators? Write a C# program to compute aggregate salary of 5 employee and then display employee record in descending order with respective employee salary using LINQ.
10. What are different type of delegate in C#? Write a C# Program to create a class 'Time' which represents time. The class should have three fields for hours, minutes and seconds. It should have constructor to initialize hours, minutes and seconds and method Time() to print current time. overload following operators a) +(Add

two time objects of 24 hours clock)b) Compare two time objects

- 11.** What is difference between `executeScalar()` and `executeReader()` method? Write a C# program to connect database Bank and insert 5 customer record (Account_no, Name, Address, Balance) in Customer table and Display the customer record whose balance is greater than 5000.